



3.2 - Mitral Regurgitation & Mitral Valve Prolapse SOAP

1) Champagne regurgitating out of diver wearing mitre hat:

Mitral regurgitation = backflow of blood from LV into LA during systole

2) Grandfather clock:

Chronic MR is more common, progressive over years/decades before symptoms manifest

3) Alarm clock:

Acute MR evolves over hours/days, with symptoms developing quickly due to lack of time for heart to adapt to hemodynamic changes

4(S) Heart Shock Jump School:

Patients with acute **MR** most commonly present in **cardiogenic shock (Chief Complaint)**

5(S) Jumper blowing puff of air:

Acute **dyspnea** = most common presenting symptom of acute **MR (Chief Complaint)**

6(S) Broken heart strings:

Ischemic acute **MR** is caused by **papillary muscle rupture**, a mechanical complication of acute MI (**HPI**)

7(S) 3 suns:

Papillary muscle rupture typically occurs **3-5 days** after MI (**HPI**)

8(S) Cut parachute strings:

Nonischemic acute **MR** is caused by **chordae tendineae rupture** (e.g., 2/2 myxomatous disease, infective endocarditis, rheumatic heart disease) (**PMHx**)

9) Hand mixer:

Myxomatous degeneration can result in chordae tendineae rupture & acute **MR**

10) Heart lantern:

Infective endocarditis (i.e., with Staph aureus) can result in chordae tendineae rupture & acute **MR**

11) Rhubarb pie near ruptured strings:

Rheumatic heart disease can result in chordae tendineae rupture & acute **MR**

12(O) BP cuff on fainting jumper:

Acute **MR** commonly presents with **hypotension** 2/2 cardiogenic shock (**Vitals**)

13(O) Crackly velcro:

In patients with acute **MR**, listen for **bibasilar crackles** 2/2 cardiogenic pulmonary edema (**Physical Exam**)

14(O) Spray of champagne:

Murmur of acute **MR** = soft, low-pitched **mid-systolic murmur** (**Cardiac Auscultation**)

15(O) Earplugs:

Many patients with severe acute **MR** have no audible murmur (i.e., “**silent MR**”) (**Cardiac Auscultation**)

16(O) Megaphone near regurgitating diver:

Echocardiography = test of choice for diagnosis of **MR** (**Echo**)

17) Acute MR DDx Clipboard:

For further workup of Acute **MR**, see Acute Dyspnea DDx

18(P) Dilated red sleeves with ivy:

IV vasodilators (e.g., nitroprusside) can reduce **MR** by decreasing SVR and promoting forward cardiac output → utility is limited by systemic hypotension (**1**)

19(P) Mitre hat rack with scalpel pegs:

Mitral valve surgery = definitive treatment for acute **MR** (**2**)

20(S) Captain blowing puff of air:

Exertional dyspnea = most common presenting symptom of chronic **MR** (**Chief Complaint**)

21(S) Heart-shaped pin with wings:

AFib is a common sequela of chronic **MR** → patients often present with **palpitations** (**Chief Complaint**)

22) 1° Pilot:

Chronic **PRIMARY MR** = disease involving mitral valve itself (e.g., leaflets, papillary muscles, chordae tendineae, annulus)

23) 2° Pilot:

Chronic **SECONDARY MR** = cardiac disease leading to regurgitation through anatomically normal mitral valve

24(S) MVP parachuter:

Mitral valve prolapse = most common cause of chronic **MR** in developed countries (**PMHx**)

25(S) Rhubarb pie near chronic clock:

Rheumatic heart disease = most common cause of chronic **MR** in the developing world → most commonly occurs in children & young adults (**PMHx**)

26(S) Red coronary crown 2° pilot hat:

Coronary artery disease = most common cause of chronic SECONDARY **MR** (**PMHx**)

27(O) Laterally displaced heart pendant:

Patients with chronic **MR** often have a **laterally displaced PMI** due to LV dilation (**Physical Exam**)

28(O) Murmur lines:

Classic murmur of chronic **MR** = blowing holosystolic murmur best heard at the apex, radiating to the axilla (**Cardiac Auscultation**)

29(O) Blowing crack in windshield:

Classic murmur of chronic **MR** = **blowing** holosystolic murmur best heard at the apex, radiating to the axilla (**Cardiac Auscultation**)

30(O) Hologram pilot's license:

Classic murmur of chronic **MR** = blowing **holosystolic** murmur best heard at the apex, radiating to the axilla (**Cardiac Auscultation**)

31(O) Keep Steady:

MR murmur **does NOT vary** with respiration (**Cardiac Auscultation**)

32(O) Squatting flight attendant gripping clipboard:

MR murmur increases with maneuvers that increase SVR (**e.g., squatting, sustained handgrip**) (**Cardiac Auscultation**)

33(O) AFib on monitor:

Left atrial enlargement caused by chronic **MR** can lead to **AFib** → look for irregularly irregular rhythm and absence of P waves on ECG (**ECG**)

34(O) Notched P-shaped cinnamon roll:

Left atrial enlargement caused by chronic **MR** can result in a broad, **notched P wave** in lead II known as “P mitrale” (**ECG**)

35(O) 50% red suit:

Regurgitant fraction >50% on echo is one definition of severe **MR (Echo)**

36) Chronic MR DDx Clipboard:

For further workup of Chronic **MR**, see Chronic Dyspnea DDx

37(P) Hat rack with scalpel pegs:

Mitral valve surgery is the definitive treatment for most patients with severe chronic PRIMARY **MR** and symptoms or evidence of LV systolic dysfunction (1)

38(P) Bandaged floppy heart balloon:

HF therapy (i.e., ACE inhibitors, beta blockers, aldosterone antagonists, diuretics, pacemaker) is the mainstay of treatment for chronic SECONDARY MR (2)

39(P) Hat rack with scalpel pegs:

Mitral valve surgery is recommended for patients with severe chronic SECONDARY **MR** who have persistent symptoms despite optimal HF therapy OR are undergoing other cardiac surgery (3)

40(S) MVP Jump Club:

MVP syndrome = controversial collection of nonspecific symptoms (e.g., chest pain, palpitations, dyspnea, anxiety, dizziness) attributed to MVP (**Chief Complaint**)

41(S) Marfan the Martian:

MVP is more prevalent in patients with **Marfan syndrome**, Ehlers-Danlos syndrome, & osteogenesis imperfecta (**PMHx**)

42(S) Loose screw:

MVP is more prevalent in patients with Marfan syndrome, **Ehlers-Danlos syndrome**, & osteogenesis imperfecta (**PMHx**)

43(S) Blue eyes:

MVP is more prevalent in patients with Marfan syndrome, Ehlers-Danlos syndrome, & **osteogenesis imperfecta** (**PMHx**)

44(O) Mid-rope CLICK!:

Classic finding of **MVP** = high-pitched **mid-systolic click** (**Cardiac Auscultation**)

45(O) Systolic spray:

MR murmur (i.e., **systolic murmur** best heard at the apex) is often detected in patients with **MVP** (**Cardiac Auscultation**)

46(O) Laughing squatting jumper:

MR murmur in **MVP** **DECREASES** with squatting (**Cardiac Auscultation**)

47(O) MVP megaphone:

Echo = diagnostic test of choice for **MVP** (**Echo**)

48(P) You can do it!:

Unless patients with **MVP** have concurrent symptoms/signs of severe chronic **MR**, **provide reassurance** that MVP is generally a benign condition (**1**)